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Cite this: *Phys. Chem. Chem. Phys.*, 2011, **13**, 11644–11656

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PAPER

Spanning QTAIM topology phase diagrams of water isomers W_4, W_5 and W_6

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Received 15th January 2011, Accepted 18th April 2011
DOI: 10.1039/c1cp20135b

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Published on 20 May 2011 on http://pubs.rsc.org | doi:10.1039/C1CP20135B

Structural and chemical properties of the small water clusters W_4, W_5 and W_6 are investigated with the theory of atoms and molecules (QTAIM). For the W_4, W_5 and W_6 clusters, nine, fourteen and twenty-seven conformers, respectively, have been analyzed. For the W_4, W_5 and W_6 clusters one, two and three of these structures, respectively, have not been reported before. We then proceed to extend the W_4, W_5 and W_6 water cluster topology space using QTAIM; the Poincaré–Hopf topological sum rules are applied to create rules to identify the spanning set of conformer topologies, this includes finding three, ten and eight new distinct topologies that satisfy the Poincaré–Hopf relation for W_4, W_5 and W_6 respectively. The topological stability of degenerate solutions to the Poincaré–Hopf relation is compared by evaluating the proximity to rupturing of critical points of the gradient vector field of the charge density. We introduce a QTAIM topology space to replace the inconsistent use of Euclidean geometry to determine whether a cluster is 1-, 2- or 3-D. We show from the topology of the charge density that the conformers of the W_4, W_5 clusters are more energetically stable in less compact, planar forms, conversely the conformers of W_6 are more energetically stable with compact 3-D topologies. Quantifying the degree of covalent character in the hydrogen bonding for the W_4, W_5 and W_6 clusters independently verifies this finding. Differences in simple rules for the number of hydrogen bonds obeying the Bernal–Fowler ice rules between W_4, W_5 and W_6 reflect the transition from 2-D to 3-D structures being more energetically stable. In addition, we identify a new class of O–O bonding interactions that are up to 48% longer than the inter-nuclear separation and appear to be failed hydrogen bonds.

I. Introduction

Copious amounts of scientific literature are routinely devoted to the study of water, arguably among the most important substances and the most studied by science.[1] The significance of water is apparent in areas as diverse as biology, chemistry, physics, astronomy, engineering, sociology, economy, politics, and most prominent, life itself: water is an essential component of living organisms and the medium in which all known biochemical processes occur. Deciphering the many unusual properties of water poses a challenge to modern science and technology, but at the same time, the unusual properties allow water to be used in a multitude of applications and maintain it as a fresh source of scientific knowledge.

Many of the macroscopically unusual properties of water (high freezing point, high boiling point, wide range of temperatures and pressures for the liquid state, lower density of ice compared to liquid water, extremely high heat capacities and latent heats, high surface tension, *etc.*) are thought to arise from behavior at the microscopic scale, therefore, detailed knowledge of the structure and properties of water in the molecular regime is needed; unfortunately, this knowledge, in many aspects, is not yet available.

Hydrogen bonding is a complex interaction with its own unresolved issues and extensive literature, it is a problem intimately connected to the structure and properties of water. The structure of liquid water is not known, many researchers describe liquid water as a collection of hydrogen-bonded clusters instead of an isotropic medium.[2–7] In an authoritative work, Buckingham and coworkers state, "Relative to chemical bonds, the hydrogen bond is weak, but in this weakness lies its importance."[8] The comparative weakness of hydrogen bonds relative to chemical bonds allows their easy formation and breaking, leading to rich conformational spaces for systems stabilized *via* hydrogen-bonding networks (HBNs), as a consequence, clusters of varying structures and sizes contribute to

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the liquid and gas phases of such systems. Complicated dynamics for HBNs rearrangements further makes difficult the characterization of these already complex conformational spaces.[9–11] Solvation processes and chemical reactions are also strongly influenced by hydrogen bonds.

Saykally and coworkers have pioneered the experimental detection of very small water clusters.[9–11] The amount of work needed for complete experimental or theoretical characterization of the potential energy surfaces (PESs) of water clusters, even for very small sizes, renders this aspiration impractical. For example, for the water hexamer, for the cage isomer alone, 14 different structures were located at the MP2/DZP level,[12] while 27 local minima were predicted at the PM3 level.[13] The studies by Restrepo and coworkers (W_4, 8 structures;[14] W5, 12 geometrical motifs, 44 structures;[15] W_6, 24 structural isomers)[16] present the largest structural variety at the B3LYP, MP2 levels. For these clusters, a large number of contributing structures are predicted: at least 3 geometrical motifs within 1 kcal mol^{-1} for W_4, at least 5 geometrical motifs within 3 kcal mol^{-1} for W5 and at least 15 geometrical motifs within 3 kcal mol^{-1} for W6 were reported. Regarding structural issues, W6 seems to be the transition point from planar to compact geometrical preferences.[16] It is interesting to notice that for most homonuclear metallic clusters, M_n, $n = 6$ is also the transition point from planar to compact structural preferences.[17–20]

Many aspects concerning hydrogen bonding remain unclear,[21] probably due in part to the difficulties in accurate theoretical treatment of the complex energy decomposition schemes for these interactions.[22] Some authors have suggested that cooperative polarization effects play a major part in the hydrogen bond interaction,[23] while others assign that role to cooperative charge transfer.[24–26] It has been suggested that dipole–dipole interactions are at play in stabilizing small hydrogen-bonded clusters.[9–11,27,28] In general, for all types of hydrogen bonds, the wide range of stabilization energies suggests contributions from all sorts of interactions, including among others electrostatic, induction, electron delocalisation, exchange repulsion, dispersion.

II. Theory and computational methods

QTAIM[29] is an extension of quantum mechanics to subdomains which define an atom as an open system and provides a powerful method for the study of bonding using only the charge density distribution $\rho(r)$. We can identify critical points in the charge density distribution where $\nabla\rho(r) = 0$, and further classify these points according to the properties of the Hessian matrix (the matrix of partial second derivatives of $F(r)$ with respect to the components of r) evaluated at these points. Diagonalizing this matrix gives the coordinate invariant (ordered) eigenvalues $\lambda_1 < \lambda_2 < \lambda_3$, and the critical points are conventionally labeled using the notation (ω, σ) where ω is the rank (the number of distinct eigenvalues) and σ is the signature (the algebraic sum of the signs of the eigenvalues). In three dimensions, there are four types of topologically stable critical points; these are denoted as (3, −3) [a local maximum, usually corresponding to a nuclear position (NCP)], (3, −1) and (3, +1) [saddle points, called bond critical points (BCPs) and ring critical points (RCPs), respectively], and (3, +3) [cage critical points (CCPs)]. We also define a piecewise continuous gradient path by evaluating $\nabla\rho$ at some point, then following this vector for an extremely small distance and evaluating $\nabla\rho$ again. In the limit of an infinitely small step, the path is continuous and corresponds to the true gradient path. The pair of special gradient paths linking a BCP with two nuclei and along which ρ is a maximum with respect to any neighboring path is known as an atomic interaction line (AIL). A bond-path is then said to exist between two nuclei linked by an AIL when the forces on the nuclei vanish (*i.e.*, the structure is in equilibrium or geometrically optimized). QTAIM therefore provides a universal definition of what constitutes bonding; if a bond-path exists between two nuclei, they are considered to be bonded. Bond-paths are usually linear (1-D), corresponding to the traditional concept of a bond, but in electron-deficient structures and/or easily ruptured bonds, they may be significantly curved.[30] Distortions of the bond-path away from linearity occur first in the direction of λ_2[30] then λ_1.

We are interested in properties evaluated at the BCPs, RCPs, and CCPs, *e.g.*, the charge density evaluated at the BCP denoted by $\rho(r_b)$ and the Laplacian $\nabla^2\rho(r_b)$ also evaluated at the BCP. Closed shell interactions (*e.g.*, ionic bonds and hydrogen bonds) are characterized by positive values of $\nabla^2\rho(r_b)$, low $\rho(r_b)$ values (<0.1 atomic units), and values of $|\lambda_1|/\lambda_3 < 1$; these types of interactions are dominated by the contraction of charge away from the BCP toward each of the nuclei. Conversely, shared interactions (*e.g.*, covalent bonds) have negative $\nabla^2\rho(r_b)$ values, high values of $\rho(r_b)$, and values of $|\lambda_1|/\lambda_3 > 1$. We can also define an ellipticity $\varepsilon = |\lambda_1|/\lambda_2 - 1$ as a measure of the relative accumulation of charge in the two directions perpendicular to the bond-path at a BCP. The conformers we are concerned with in this work are molecular and the numbers of the four different types of three dimensional critical points are related by a fundamental theorem of topology; the Poincaré–Hopf relationship:[29]

$$n - b + r - c = 1 \qquad (1)$$

where n, b, r, and c are the numbers of NCPs BCPs, RCPs, and CCPs, respectively. All care must be taken to ensure that all of the critical points are found since it is possible that if a critical point finding algorithm misses both a BCP and an RCP from the same charge density distribution then an apparently valid solution of (1) is possible. The sum Σ_{BRC} provides us with a measure of the overall topological complexity of a structure; the higher the sum, the higher the complexity. We can use eqn (1) to explore a conformer phase space by plotting the number of RCPs against the number of BCPs. The phase space will be bounded by structures whose topologies have been determined either previously or during the course of this work, this reduces the degree of conjecture in the work. In this way we can clearly identify the topologies of any missing conformers as existing in a diagonal band amongst previously identified structures. The thickness of this band is determined by the number of possible topologically stable CCPs, which is in turn indicated by the number of CCPs that would exist for the most compact (3-D) structure, such as a regular polyhedron with the given number of NCPs.

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For consistency, in this work for the concept of quantum topology (QT) within QTAIM rather than Euclidean geometry will be used to determine whether a structure is 3-D, 2-D or 1-D., We shall use the following notation to refer to all possible molecular geometries consistent with QTAIM: 3-D_{QT}, 2-D_{QT} or 1-D_{QT}. To the authors knowledge they are unaware of any other work in the literature taking this approach. The presence of a CCP is a necessary condition for a structure to be considered to be quantum topologically 3-D_{QT}. A structure containing a CCP possesses a region that is usually completely bounded by bond-paths with an RCP in the plane of each enclosing face. A rare exception to this is the first example of a CCP in a single RCP contained in a twisted helical topology.[31] Structures can possess 3-D Euclidean geometry but without the presence of a CCP these structures are not considered to be 3-D quantum topological objects. Such structures without CCPs do not contain an enclosed region in the topology of the gradient of the charge density, but instead have an 'open box' form with no 'lid' and are 2-D_{QT} quantum topological objects. Structures without CCPs but with RCPs are determined to be 2-D_{QT} and structures with neither CCPs nor RCPs are 1-D_{QT}. In this work all structures with a CCP are 3-D_{QT} and all other structures are 2-D_{QT} where the dimensionality D_{QT} always refers to a quantum topology defined within QTAIM and not Euclidean geometry.

In order to determine the topological stability of the CCPs we adopt the recently used measure D_{rc} which is simply the separation between a CCP and the closest RCP. In order to determine the degree of strain in a bond-path we calculate $BPL - GBL - I$, which is the difference between the bond-path length and the distance between the nuclear attractors. Previously, it was determined that bond-paths can be linear (1-D), 2-D or 3-D.[30] In conformers where O–O BCPs exist we will use the relation D_{rbO-O}, which represents the spatial separation between an O–O BCP and the closest RCP. We do not regard the O–O BCPs as chemical bonds,[32] but rather as indicators of the topological instability of a structure and the proximity of nearby conformers on the PES.[33–37] Strictly, we should use the separation between the longer RCP–BCP path which approximates the geometric distance for short BCP–RCP separation.

In order to address the issue of cooperativity in hydrogen bonding we use the local total energy density at the BCP suggested by[38]

$$H(r_{bcp}) = V(r_{bcp}) + G(r_{bcp}) \qquad (2)$$

Previously, one of us found that the usual strength of hydrogen bonds in ice Ih could be explained in terms of eqn (2),[31] in agreement with the experiment,[39] where a degree of covalent character is indicated by $H(r_{bcp}) < 0$. All of the QTAIM properties were calculated using AIMQB program within the AIMStudio suite,[40] using the Proaim basin integration method.

Most cluster geometries were extracted from previous work on the W_4, W_5, W_6 systems.[9–11] Those structures were obtained after random walks of the PM3 potential energy surfaces under annealing conditions, followed by geometry optimizations at the B3LYP/6-311++G** level and further MP2/6-311++G** refinement and characterization as true minima (see references for details). The structures that are needed to completely span the topological phase diagrams were optimized and characterized by counting the number of negative eigenvalues of the Hessian matrix at the MP2/6-311++G** level, followed by highly correlated CCSD(T) energy calculations.

III. Results and discussion

In Fig. 1–3 the critical point sets and the RCP–BCP paths of the topologies of the charge density of a selection of the conformers of W_4, W_5 and W_6, respectively, from eqn (1) are presented. In Fig. 1–3 the order of appearance of the conformer topology diagrams corresponds to the ordering in Tables 1–3 respectively. Also in Fig. 1–3 the solid hydrogen bond bond-paths indicate that there is a degree of covalent character present in the hydrogen bond as indicated by $H(r_{bcp}) < 0$ in eqn (2). The converse is true for the hydrogen bonds represented by dashed lines. In Fig. 4(a)–(c) the phase diagrams of the QTAIM topologies of W_5 and W_6 are shown. The degree of covalent character $H(r_{bcp})$, from eqn (2), is plotted against the bond-path length (BPL) for each of the three isomer sets W_4, W_5 and W_6, respectively, in Fig. 5(a)–(c). In Tables 1–3 inclusive we present the results for the Poincaré–Hopf relationship for W_4, W_5 and W_6 in the order of increasing numbers of BCPs and the 'missing' topologies predicted by the Poincaré–Hopf relation. In Tables 4–6 we presented the point properties of the W_4–W_6 conformers that relate to the energetic stability of the conformers. In this study, string-like conformers, which correspond to valid solutions of the Poincaré–Hopf relation with $BCP = NCP - 1$, were not included as this was a study concerning clusters. In this study we used the Poincaré–Hopf relation to create one more structure for W_4; conformer IX (oxygen atoms in the corners of a regular tetrahedron), two structures for W_5: *PYRAMID_5A* and *PYRAMID_5B* (oxygen atoms at the vertices of trigonal bi-pyramids), and three structures for W_6; *PYRAMID_A*, *PYRAMID_B* and *PYRAMID_C* (oxygen atoms at the vertices or square bi-pyramids). All of the pyramid structures were constructed to obey the Bernal–Fowler ice rules before performing geometry optimizations. All pyramid structures were predicted to be saddle points at the MP2/6-311++G** level, except for the W_4 pyramid, which is a true minimum.

(i) The Poincaré–Hopf relation to generate a phase space of isomer topologies

We begin by examining the topologies of the nine tetramers of the W_4 isomers given in Table 1, where it can be seen that there are 5 distinct solutions for the Poincaré–Hopf relation and 4 distinct values of the sum Σ_{BRC}, of eqn (1). Entries in bold denote structures calculated in this work where the topological analysis for all of the structures in this study is entirely new. From Table 1 and Fig. 4(a) we can see that an additional 3 topologies, *A*–*C*, for W_4 are described which will correspond to either global, local minima or saddle-points *i.e.* transition states on the potential energy surface (PES). It is highly unlikely that any of the topologies *A*–*C* are global minima, but it is possible that they are local minima and given that we know the structures with the topologies *A*–*C* are allowed in

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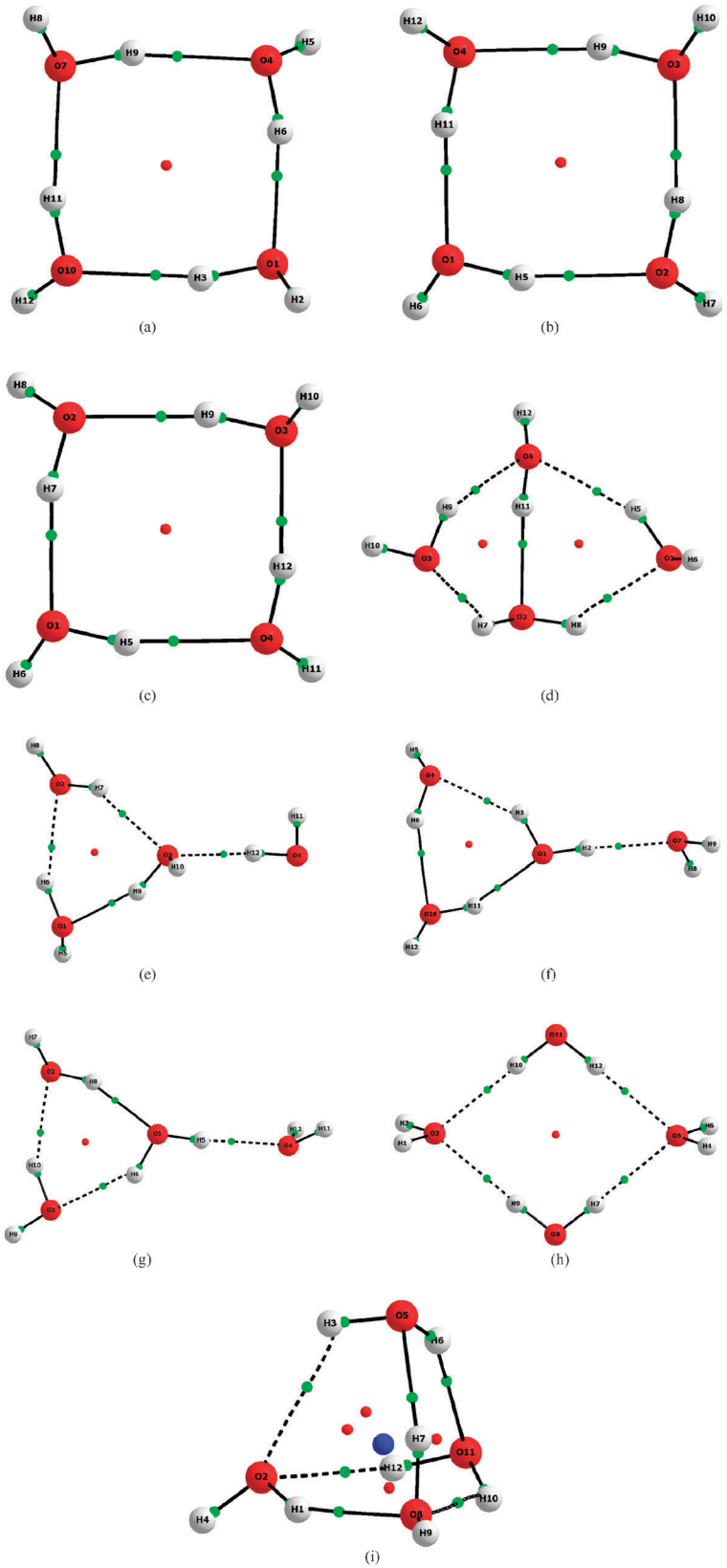

Fig. 1 The critical point sets, BCPs, RCPs and CCPs, of the topologies of the charge density of the conformers of W_4. The BCPs, RCPs and CCPs are visible as unmarked green, red and blue spheres, respectively, in the electronic version. The relative sizes of the critical points are as follows: BCP < RCP < CCP. In sub-figures (a)–(i) the topologies of conformers I–IX are shown (see Table 1). The structure for W_4 was created in this work, conformer IX is presented in sub-figure (i). The hydrogen bond bond-paths represented by solid lines possess a degree of covalent character, *i.e.*, $H(r_{bcp}) < 0$, from eqn (2), the converse is true for the dashed hydrogen bond bond-paths. This also holds true for Fig. 2 and 3.

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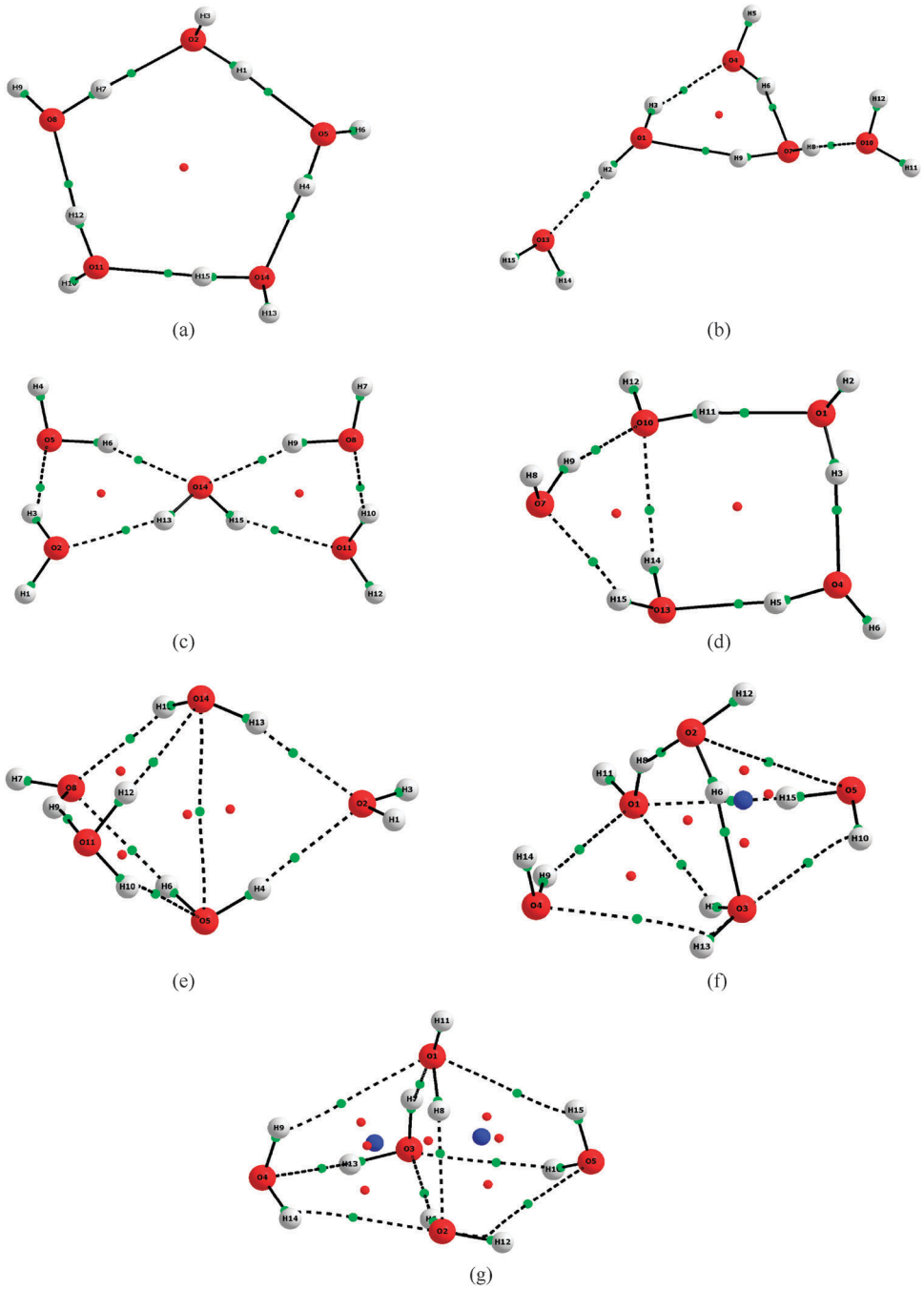

Fig. 2 The critical point sets for the conformers of W_5, see also the caption of Fig. 1. In sub-figures (a)–(d) the topologies of the *PUCKERED_RING*, *TRIMER_P1_P1_TRANS*, *DOUBLE_TRIMER* and *TETRAMER_TRIMER* W_5 conformers, respectively, are presented. The CAGE conformer is presented in sub-figure (e), notice the lack of CCPs. The two W_5 structures *PYRAMID_5A* and *PYRAMID_5B* created for this work are presented in sub-figures (f) and (g) respectively.

terms of satisfying both the Bernal–Fowler ice rules[41] and the Poincaré–Hopf relation then *A*–*C* correspond to structures that will at least be transition states. One of the topologies of W_4, IX, (see Fig. 1(i)) a heavy-atom tetragonal structure, has the same topology that has previously been published for a W_4 cluster, where it is referred to as a prism structure, see Fig. 1 in the work of Shields,[42] and it is the most compact(3-D_{QT}) W_4 structure possible and so forms the uppermost point of our topology phase diagram (see Fig. 4(a)), notice the CCP inside the structure. The same is true for the new topologies *A*–*J* in Table 2 and Fig. 4(b) for W_5 and for *A*–*H* in Table 3 and Fig. 4(c) corresponding to W_6. In Tables 2 and 3, which refer to the sets of topologies for W_4, W_5 and W_6, the most complex structures, that is, *IX*, *PYRAMID_5A*–*B* and *PYRAMID_6A*–*C* (see Fig. 1(i), 2(f) and (g) and 3(j)–(l) respectively), are all heavy-atom regular polyhedra, tetrahedron, trigonal and square bi-pyramids. Conformer IX of W_4 obeys the Bernal–Fowler ice rules but none of the *PYRAMID_5A*–*B* and *PYRAMID_6A*–*C* structures do after relaxation. There are no cage critical points (CCPs) in any of the isomers I–VIII

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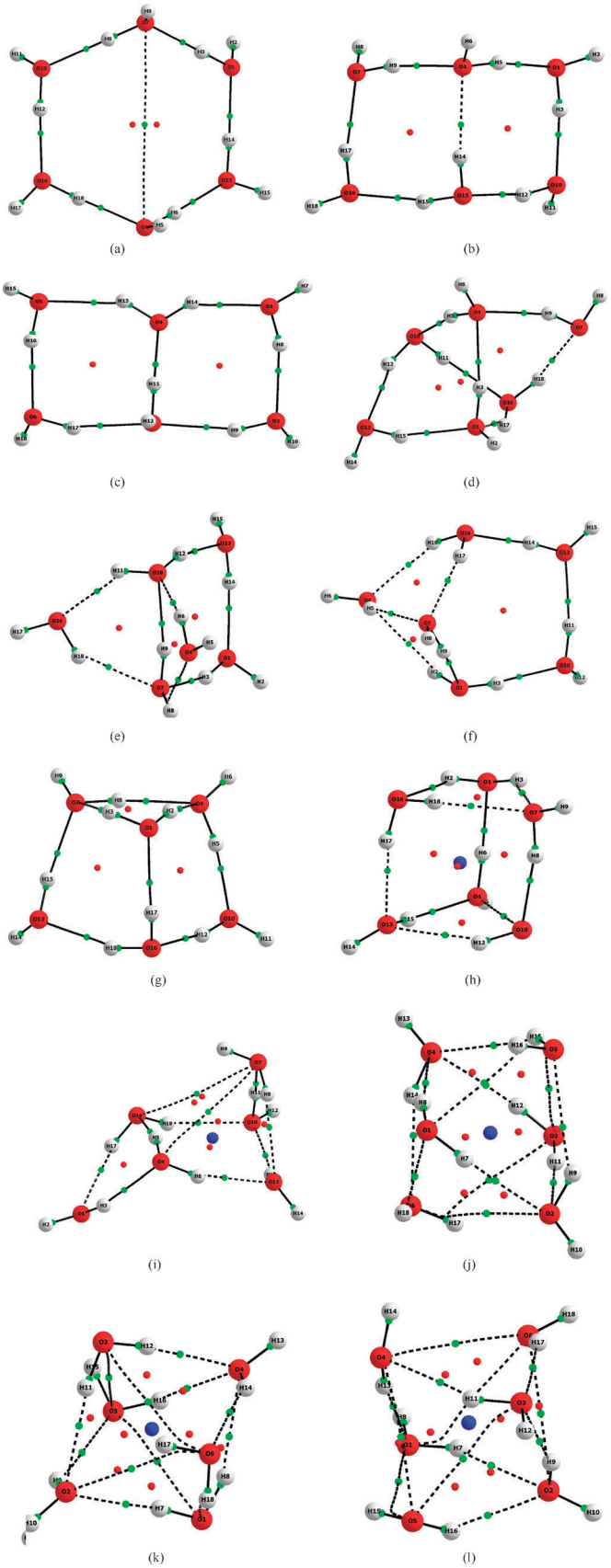

Fig. 3 The critical point sets for the conformers of W_6, see also the caption of Fig. 1. A selection of the W_6 conformers is shown in sub-figures (a)–(i), corresponding to the *BOAT*, *BOOK*, *DOUBLE_TETRAMER*, *CAGE*, *CLOSED_BAG*, *INTERMEDIATE_BAG*, *PRISM_BOOK*, *PRISM*, *CLOSED_CHAIR_DOWN* respectively. The topology of the three prism structures *PYRAMID_6A–6C* is displayed in sub-figures (j)–(l) respectively.

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Table 1 Critical point data for the three varying sets of critical points for the quadramer structures W_4, see the main text for further explanation. The number of atoms in each structure is 12 in all cases. Valid conformer types of the 'missing' Poincaré–Hopf relations are denoted by the letters A–C inclusive. ΔE CCSD(T) corresponds to the relative binding energy calculated with respect to the most energetically stable structure. All CCSD(T)/aug-cc-pvdz calculations using the MP2/6-311++g** optimized geometries. All relative energies corrected for the unscaled MP2/6-311++g** ZPEs. All energies in kcal mol^{-1}

Conformer	BCP	RCP	CCP	Σ_{BRC}	ΔE CCSD(T)
I	12	1	0	13	0.0
II	12	1	0	13	1.0
III	12	1	0	13	1.1
V	12	1	0	13	7.3
VI	12	1	0	13	7.1
VII	12	1	0	13	7.2
VIII	12	1	0	13	12.0
IV	13	2	0	15	5.9
A	**12**	**2**	**1**	15	
B	**13**	**3**	**1**	**17**	
C	**14**	**3**	**0**	**17**	
IX[a]	**14**	**4**	**1**	**19**	**15.5**

[a] Structure corresponds to a saddle-point in the energy surface.

Table 2 Critical point data for the three varying sets of critical points for the pentamer structures W_5, see the main text for further explanation. The number of atoms in each structure is 15 in all cases. The numbers in brackets under the BCP column heading refer to numbers of O–O BCPs. Valid conformer types of the 'missing' Poincaré–Hopf relations are denoted by the letters A–J inclusive. See the caption of Table 1 for more details

Conformer	BCP	RCP	CCP	Σ_{BRC}	ΔE CCSD(T)
PUCKERED_RING	15	1	0	16	0.0
ENVELOPE	15	1	0	16	1.0
TRIMER_P1_P1_TRANS	15	1	0	16	9.1
TETRAMER_P1	15	1	0	16	0.8
DOUBLE_TRIMER	16	2	0	18	4.2
TRIMER_P1_P1_CIS	15	1	0	16	9.0
TRICYCLE	16	2	0	18	1.2
TRIMER_P2	15	1	0	16	9.4
TETRAMER_TRIMER	16	2	0	18	0.8
BICYCLE_P1	16	2	0	18	6.2
CAGE	18(1)	4	0	22	5.9
BICYCLE_P1_A	16	2	0	18	6.7
A	**15**	**2**	**1**	**18**	
B	**15**	**3**	**2**	**20**	
C	**16**	**3**	**1**	**20**	
D	**16**	**4**	**2**	**22**	
E	**17**	**3**	**0**	**20**	
F	**17**	**4**	**1**	**22**	
G	**17**	**5**	**2**	**24**	
PYRAMID_5A[a]	**18(2)**	**5**	**1**	**24**	**8.5**
H	**18**	**6**	**2**	**26**	
I	**19**	**5**	**0**	**24**	
J	**19**	**6**	**1**	**26**	
PYRAMID_5B[a]	**19(1)**	**7**	**2**	**28**	**7.6**

[a] Structure corresponds to a saddle-point in the energy surface.

of W_4 as can be seen in Table 1 and Fig. 1(a)–(h) respectively, indicating that planar (2-D_{QT}) geometries are favored over compact (3-D_{QT}) ones. Isomer IX has a CCP but is the most energetically unstable structure for the W_4 energies. Neither are there any O–O BCPs in any of the energetically stable W_4 conformers due to a lack of both compactness in the structures and complexity in the PES.[43] In Tables 2 and 3 it can be seen that there is a strong tendency to favor compact (3-D_{QT})

Table 3 Critical point data for the three varying sets of critical points for the hexamer structures W_6, see the main text for further explanation. The number of atoms in each structure is 18 in all cases. The numbers in brackets under the BCP column heading refer to numbers of O–O BCPs. Valid conformer types of the 'missing' Poincaré–Hopf relations are denoted by the letters A–H inclusive. See the caption of Table 1 for more details

Conformer	BCP	RCP	CCP	Σ_{BRC}	ΔE CCSD(T)
CHAIR	18	1	0	19	1.6
TWISTED_BOAT	18	1	0	19	2.5
PENTAMER_P1	18	1	0	19	4.5
TETRAMER_P2A_P1	18	1	0	19	6.6
BOAT	19(1)	2	0	21	2.6
BOOK	19	2	0	21	0.8
DOUBLE_TETRAMER	19	2	0	21	3.0
OPEN_BAG	19	2	0	21	1.2
PENTAMER_DOWN_1	19	2	0	21	2.0
PENTAMER_PLANAR_P1	19	2	0	21	2.2
PENTAMER_UP_P1	19	2	0	21	2.8
TETRAMER_TRIMER	19	2	0	21	3.0
TRICLYCLEP_1B	19	2	0	21	4.0
TRIMER_TRIMER_P1	19	2	0	21	8.6
CAGE	20	3	0	23	1.1
CLOSED_CHAIR	20	3	0	23	6.1
CLOSED_BAG	20(1)	3	0	23	3.4
DOUBLE_ENVELOPE	20	3	0	23	2.7
INTERMEDIATE_BAG	20	3	0	23	1.9
PRISM_BOOK	20(1)	3	0	23	1.4
TRICLYCLEP_1A	20	3	0	23	3.2
TRIPLE_TRIMER	20	3	0	23	6.7
PRISM	21	5	1	27	0.0
CLOSED_CHAIR_DOWN	22(2)	6	1	29	5.5
A	**18**	**2**	**1**	**21**	
B	**19**	**3**	**1**	**23**	
C	**20**	**4**	**1**	**25**	
D	**21**	**4**	**0**	**25**	
E	**22**	**5**	**0**	**27**	
F	23	6	0	29	
G	**23**	**7**	**1**	**31**	
H	**24**	**7**	**0**	**31**	
PYRAMID_6A[a]	**24(3)**	**8**	**1**	**33**	**14.5**
PYRAMID_6B[a]	**24(4)**	**8**	**1**	**33**	**12.8**
PYRAMID_6C[a]	**24(6)**	**8**	**1**	**33**	**11.2**

[a] Structure corresponds to a saddle-point in the energy surface.

structures in W_6 over W_5, since W_5 possesses not very energetically stable conformers with CCPs and W_6 has 4, including the most energetically stable structure *PRISM* (see Fig. 3(h)).

We observe that the conformers with the highest values of topological complexity $\sum_{BRC}$ also contain O–O BCPs. Consistent with this, the most energetically stable W_4 clusters do not contain any O–O BCPs. Previously,[37] O–O BCPs have been linked to the presence of nearby conformers on the PES. An example of this is the O–O BCP in the *BOAT* conformer of W_6 (see Fig. 3(a)), where similar structures include the *BOOK* and the *DOUBLE TETRAMER* (see Fig. 3(b) and (c) respectively) in which both have the same topological complexity $\sum_{BRC}$. For the W_5 isomer set the missing topologies *A–L* may well contain structures that are clustered around the *CAGE* structure, see Fig. 3(d). Consistent with this, an O–O BCP exists for the most complex of the previously studied W_5 clusters, the *CAGE*. For the W_6 clusters the most energetically stable structure is the *PRISM* (see Fig. 3(h)), which contains a CCP but no O–O BCPs. For the W_6 structures there exist energetically stable 2-D_{QT} and 3-D_{QT} structures and the most

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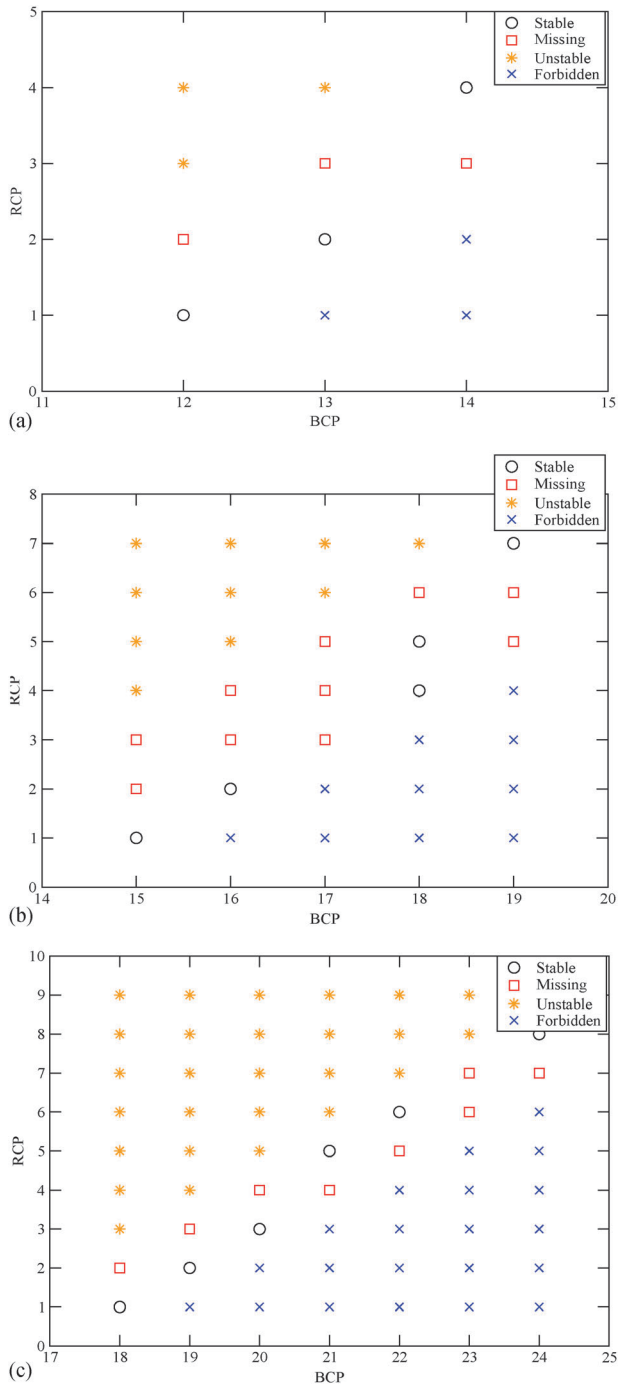

Fig. 4 Phase-diagrams of the solutions of the Poincaré–Hopf relation and the sum rules of Σ_{BRC} for W_4, W_5 and W_6 are given in sub-figures (a), (b) and (c) respectively. Permitted topologically and energetically stable solutions are denoted by black circles or red squares. The upper left plot area marked with asterisks denotes the region of topological instability based on there being increasingly more CCPs. The region marked as 'forbidden' is excluded since no valid solutions exist for the Poincaré–Hopf relation. In addition we exclude regions with no RCPs as these solutions to the Poincaré–Hopf relation correspond to string like structures as opposed to clusters.

energetically stable structure resides mid-way along the phase diagram at (21, 5), this result is in agreement with a previous study on $(SiO_2)_6$, although in the previous study the most energetically stable structure was 2-D_{QT} and not 3-D_{QT}.[44] For W_5, the structures with O–O BCPs are the more complex topologies with values of $\sum_{BRC} \geq 22$, showing that these structures are more topologically and energetically unstable than those with lower values of $\sum_{BRC}$. The conformers with O–O BCPs in the W_6 clusters exist either side of the mid-point, (21, 5), of the conformer phase diagram (see Fig. 4(c)) as well as at the mid-point. This provides evidence of the process of structural transition from 2-D_{QT} structures; none of the known structures from (18, 1) to (20, 3) contain CCPs, the transition occurs at (21, 5) with the *PRISM* structure. The Poincaré–Hopf solution of the *PYRAMID_6A–C* structures is represented by a point (24, 8) in Fig. 4(c) that marks the 'upper' boundary of the phase space for the W_6 conformers.

(ii) The relationship between the energetic stabilities of conformers and the associated BCPs, RCPs and CCPs

Within a given set of identical topologies we can compare the overall energetic stabilities of conformers by comparison of the topological stability of the associated BCPs, RCPs and CCPs. In order to do this we use the simple QTAIM relations; ellipticity, ε_{H-O}, ε_{O-O}, bond-path deviation $BPL - GBL - I$, and critical point separation, D_{rc} and D_{rbO-O} as outlined in Section II. From Table 1 it can be seen that the W_4 conformers I–VIII, excluding IV, have identical solutions to the Poincaré–Hopf relation, see also Fig. 1(a)–(h) respectively, excluding Fig. 1(d). Note that there are degeneracies in the conformers produced corresponding to these solutions of the Poincaré–Hopf relation, *i.e.* see Fig. 1(a)–(c) and (e)–(g) respectively. Degeneracies in the conformers occur for the W_5 and W_6 clusters and we only include one sub-figure from each set of degenerate conformers in Fig. 2 and 3 corresponding to the solutions of the Poincaré–Hopf relation in Tables 1–3.

Using the QTAIM relations outlined above it can be seen in Table 4 that conformer I is more topologically stable than conformer II which in turn is more topologically stable than conformer III which is in agreement with the results from MP2 calculations.[9] Conformers V–VII (see Fig. 1(e)–(g) respectively) all possess the same Poincaré–Hopf relations and from Table 2 we can again separate out the less topologically stable VI and VII in agreement with MP2 from the more energetically stable conformer V by observing that V has lower values of ellipticity, ε_{H-O} and bond-path deviation $BPL - GBL - I$. The least topologically and energetically stable conformer of W_4 is structure IX, this is clearly reflected in the fact that the values of ellipticity, ε_{H-O}, and $BPL - GBL - I$ are an order of magnitude larger for IX than for conformers I–VIII. Similar effects are apparent for W_5 and W_6 in Tables 5 and 6, where for W_5 the most topologically and energetically unstable structures *PYRAMID_5A* and *PYRAMID_5B* have the largest values for ellipticity, ε_{H-O}, and bond-path deviation $BPL - GBL - I$. In addition these two conformers are the only W_5 structures to consist of O–O interactions, in addition, they also possess large values of ellipticity, ε_{O-O}, and bond-path deviation $BPL - GBL - I$. Previously, it has been observed that O–O interactions are associated with imminent structural change and flatter regions on potential energy surfaces, therefore the structures with O–O interactions are more energetically

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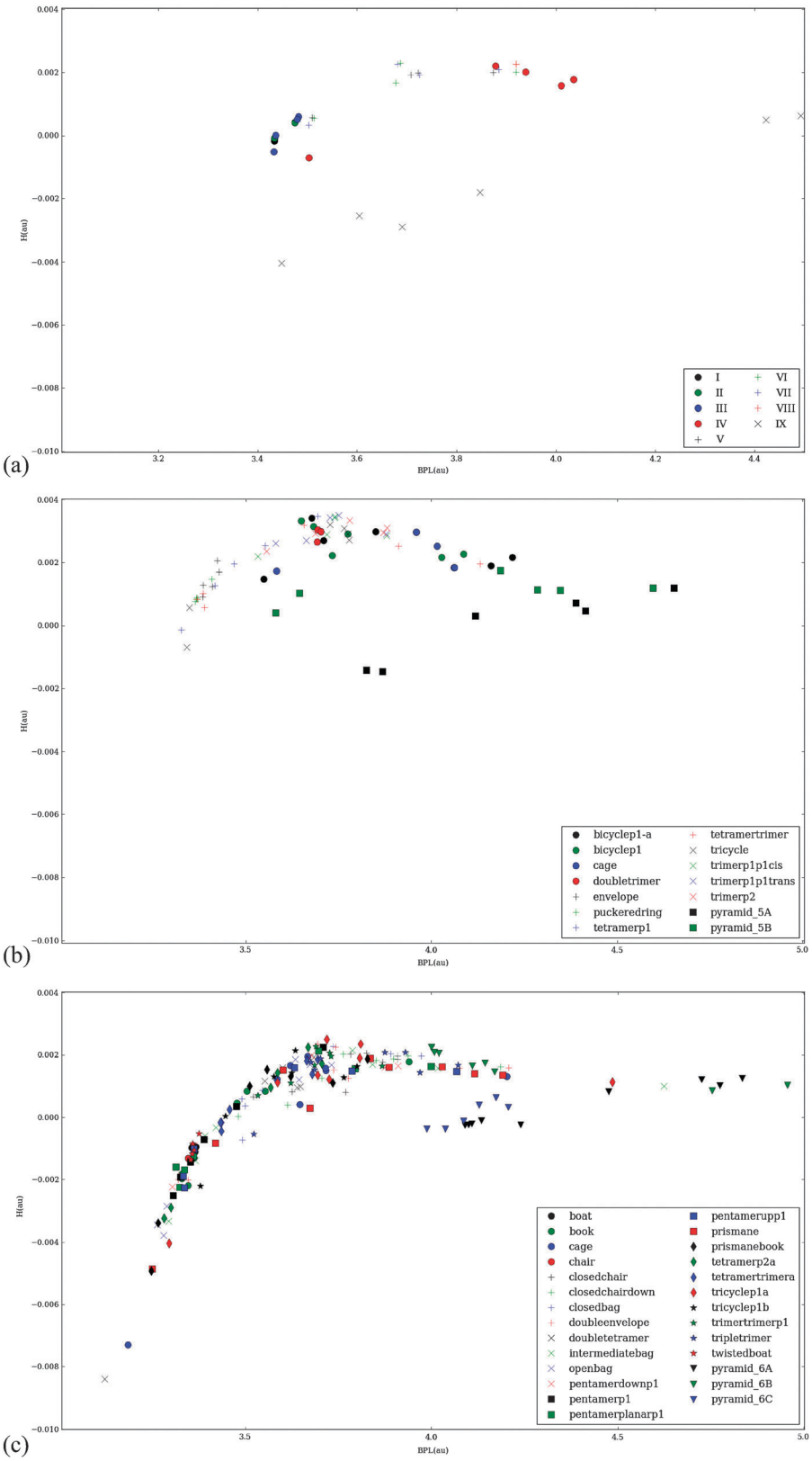

Fig. 5 The variations of the covalent bond character $H(r_{bcp})$ with BPL for each of the three cluster types W_4,W_5 and W_6 are presented in sub-figures (a), (b) and (c) respectively. See also Fig. 1–3 where the hydrogen bonds with the highest degree of covalent character are denoted by solid lines.

unstable than those without. The maximum possible combined number of hydrogen bonds and O–O BCPs that a W_6 octagonal structure could contain is 12, with numbers of hydrogen bonds being 9, 8 and 6, respectively, and the shortfall being made up with O–O BCPs. But for W_6, the octagonal conformers *PYRAMID_6A–C* with more O–O BCPs are more

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Table 4 Analysis of critical point properties of the closed shell (hydrogen bonds) conformers of $(H_2O)_{4-}$. *BPL − GBL − I* corresponds to the ratio of the shortest separation of the bonded nuclei with the *BPL* and BPL refers to the bond-path length. The scalar D_{rc} refers to the separation between a CCP and the closest RCP. All units of distance are in Bohr. The highest values of the ellipticity ε_{H-O} for the hydrogen bond BCPs are chosen for each structure

Conformer	H–O interaction	Ellipticity, ε_{H-O}	*BPL − GBL − I*	BPL	D_{rc}
I	H6–O1	0.034	0.003	3.433	
II	H5–O2	0.082	0.006	3.475	
III	H5–O4	0.085	0.006	3.479	
IV	H9–O4	0.059	0.017	3.878	
V	H7–O3	0.071	0.020	3.874	
VI	H3–O4	0.072	0.027	3.919	
VII	H10–O2	0.067	0.016	3.726	
VIII	H12–O5	0.044	0.002	3.919	
IX	**H3–O2**	**0.212**	**0.059**	**4.493**	**1.169**

Table 5 Analysis of critical point properties of the closed shell (hydrogen bond and O–O) interactions of conformers of $(H_2O)5$. The scalar D_{rbO-O} refers to the separation between an O–O BCP and the closest RCP. See the caption of Table 4 for more details

Conformer	H–O interaction	Ellipticity, ε_{H-O}	*BPL − GBL − I*	BPL	D_{rc}
PUCKERED RING	H12–O8	0.083	0.004	3.409	
ENVELOPE	H7–O5	0.096	0.004	3.423	
TRIMER_P1_P1_TRANS	H6–O7	0.061	0.012	3.581	
TETRAMER_P1	H15–O5	0.053	0.002	3.693	
DOUBLE TRIMER	H15–O11	0.060	0.018	3.703	
TRIMER_P1_P1_CIS	H9–O10	0.062	0.019	3.718	
TRICYCLE	H15–O8	0.069	0.021	3.778	
TRIMER_P2	H9–O10	0.070	0.030	3.871	
TETRAMER_TRIMER	H15–O7	0.066	0.037	3.912	
BICYCLE_P1	H10–O3	0.065	0.040	4.026	
CAGE	H12–O14	0.080	0.040	4.061	
BICYCLE_P1_a	H12–O3	0.081	0.057	4.160	
PYRAMID_5A	**H10–O3**	**0.094**	**0.102**	**4.597**	**0.824**
PYRAMID_5B	**H14–O2**	**0.375**	**0.128**	**4.653**	**0.875**
Conformer	**O–O interaction**	**Ellipticity, ε_{O-O}**	***BPL − GBL − I***	**BPL**	**D_{rbO-O}**
CAGE	O5–O14	1.547	0.018	6.716	0.436
PYRAMID_5A	**O2–O5**	**0.224**	**0.014**	**6.496**	**1.500**
	O3–O4	**0.392**	**0.479**	**6.190**	**1.939**
PYRAMID_5B	**O2–O5**	**0.535**	**0.411**	**6.077**	**0.883**

energetically stable; *PYRAMID_6A–C* possess 3, 4 and 6 O–O interactions, respectively, and their relative energetic stabilities from Table 3 are 14.5, 12.8 and 11.2 kcal mol^{-1} respectively. It can be seen for the O–O interactions in the structures that obey the Bernal–Fowler ice rules that the correlation with energetic stability and the bond-path strain, *BPL − GBL − I* is reversed; the structures that contain O–O BCPs with the most strained O–O bond-paths are the more energetically stable. An example of this is the O4–O7 BCP of the *BOAT* structure in the W_6 clusters (see Table 6 and Fig. 3(a)), the BPL is 9.3 au and was only included to keep the Poincaré–Hopf sum relations consistent (see Table 3), showing that this sole O–O BCP is very topologically unstable and not surprisingly the value of the ellipticity is extremely high (> 1.0). For the remaining W_6 structures, *i.e.* the ones that obey the Bernal–Fowler ice rules, the O–O BCPs in the order of both increasing energetic stability and *BPL − GBL − I* are *CLOSED_CHAIR_DOWN*, *CLOSED_BAG* and *PRISM_BOOK* (see Fig. 3(i), (e) and (g) respectively). This finding again demonstrates the correlation between energetic stability and O–O BCP instability. For the *PRISM_BOOK* structure the value of *BPL − GBL − I* (= 0.48) is extremely high and is the highest of all the O–O BCPs of this study, the *PRISM_BOOK* structure is the fifth most energetically stable structure for W_6. This finding is not apparent for the *PYRAMID_6A–C* structures none of which obey the Bernal–Fowler ice rules where the order of topological stability increases with the number of O–O interactions as can be seen from Tables 3 and 6. The *PYRAMID_5A–B* structures are more energetically stable with less O–O interactions in contrast to the *PYRAMID_6A–C*, this inconsistency can be explained by the preference for W_6 clusters to possess more complex topologies than W_5 or W_4 conformer sets, the latter doesn't contain any O–O interactions.

From Tables 1–3 it can be seen that the 3 conformer sets of this study, W_4, W_5 and W_6, all contain CCPs. The W_4 and W_5 conformers with CCPs, *IX* and *PYRAMID_5A* and *PYRAMID_5B*, respectively, are very energetically unstable, as can be seen by a comparison of the relative energies from CCSD(T) calculations in Tables 1 and 2. Conversely, for W_6 the most energetically stable structure, *PRISM*, see Table 3, contains a CCP. The W_6 conformer set contains an additional 4 structures with CCPs: *CLOSED_CHAIR_DOWN*, *PYRAMID_6A*, *PYRAMID_6B* and *PYRAMID_6C*. From Table 6, it can be seen that the first of these structures *CLOSED_CHAIR_DOWN* possesses much lower values of D_{rc} than the *PRISM* and so is much less topologically stable.

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Table 6 Analysis of critical point properties of the closed shell (hydrogen bond and O–O) interactions of conformers of $(H_2O)_6$, see the captions of Tables 4 and 5 for more details

Conformer	H–O interaction	Ellipticity, ε_{H-O}	$BPL - GBL - I$	BPL	D_{rc}
CHAIR	H2–O4	0.042	0.001	3.346	
TWISTED_BOAT	H12–O13	0.075	0.002	3.362	
PENTAMER_P1	H18–O2	0.082	0.002	3.306	
BOAT	H12–O6	0.063	0.002	3.356	
TETRAMER_P2A_P1	H14–O5	0.052	0.002	3.668	
BOOK	H5–O1	0.077	0.003	3.329	
OPEN_BAG	H17–O10	0.083	0.002	3.673	
PRISM_BOOK	H15–O7	0.076	0.002	3.558	
TETRAMER_TRIMER	H6–O7	0.077	0.005	3.458	
DOUBLE_TETRAMER	H8–O2	0.088	0.005	3.600	
TRICLYCLEP_1B	H12–O13	0.073	0.004	3.624	
TRIMER_TRIMER_P1	H17–O4	0.068	0.019	3.726	
CLOSED_CHAIR	H8–O4	0.088	0.009	3.868	
PENTAMER_UP_P1	H8–O6	0.083	0.002	3.361	
CAGE	H15–O1	0.085	0.004	3.667	
PENTAMER_DOWN1	H11–O16	0.073	0.005	3.677	
CLOSED_BAG	H6–O10	0.112	0.010	3.735	
PENTAMER_PLANAR_P1	H16–O6	0.084	0.002	3.361	
PRISM	H18–O7	0.103	0.023	4.027	1.464
TRIPLE_TRIMER	H5–O13	0.065	0.016	3.702	
CLOSED_CHAIR_DOWN	H6–O13	0.104	0.010	3.852	0.477
DOUBLE_ENVELOPE	H14–O7	0.072	0.006	3.742	
INTERMEDIATE_BAG	H2–O4	0.078	0.085	4.623	
TRICLYCLEP_1A	H15–O7	0.149	0.157	4.485	
PYRAMID_6A	**H16–O1**	**0.315**	**0.158**	**4.836**	**2.465**
PYRAMID_6B	**H18–O1**	**0.146**	**0.312**	**4.956**	**2.382**
PYRAMID_6C	**H17–O3**	**0.122**	**0.014**	**4.173**	**2.321**
Conformer	O–O interaction	Ellipticity, ε_{O-O}	$BPL - GBL - I$	BPL	D_{rbO-O}
BOAT	O4–O7	1.062	0.011	9.293	0.550
CLOSED_CHAIR_DOWN	O7–O16	0.858	0.039	6.665	0.511
	O4–O7	2.555	0.074	6.713	0.641
CLOSED_BAG	O4–O7	0.741	0.429	6.010	1.610
PRISM_BOOK	O1–O7	0.378	0.477	6.526	1.301
PYRAMID_6A	**O3–O6**	**0.543**	**0.450**	**6.617**	**1.480**
	O2–O6	**0.639**	**0.364**	**6.531**	**1.355**
	O3–O5	**1.286**	**0.127**	**6.295**	**1.224**
PYRAMID_6B	**O3–O5**	**0.226**	**0.012**	**6.141**	**1.422**
	O2–O6	**0.610**	**0.273**	**6.402**	**1.404**
	O3–O6	**0.903**	**0.213**	**6.342**	**1.530**
	O1–O5	**2.524**	**0.047**	**5.802**	**1.574**
PYRAMID_6C	**O4–O6**	**0.121**	**0.021**	**6.094**	**1.535**
	O2–O6	**0.237**	**0.017**	**5.720**	**1.614**
	O3–O5	**0.422**	**0.064**	**6.138**	**1.543**
	O1–O5	**0.427**	**0.334**	**6.037**	**1.746**
	O4–O5	**0.448**	**0.031**	**6.105**	**1.253**
	O1–O6	**0.786**	**0.245**	**6.319**	**1.305**

The *PYRAMID A–C* structures also have large D_{rc} which at first glance might appear to be more topologically stable than the *PRISM* structure of W_6 until it is realized that the CCPs only exist in the *PYRAMID A–C* structures by virtue of O–O BCPs, which are much weaker and more susceptible to rupture than hydrogen BCPs. The most energetically stable W_6 structure, *PRISM*, does not contain any O–O BCPs, unlike the more energetically unstable *PYRAMID_6A*, *PYRAMID_6B* and *PYRAMID_6C*.

It can be noticed from sub-Fig. 3(e) and (g) that some of the bond-paths of the associated O–O BCPs in W_6 are very distorted; for instance, the bond-paths O4–O7 and O1–O7 in the *CLOSED_BAG* and *PRISM_BOOK* structures respectively. This distortion is apparent for example, in the O1–O7 bond-path in the *PRISM_BOOK* structure, where bond-path originates from O7, then skims past the hydrogen atom, narrowly avoiding it before sharply turning and terminating at O1, the degree of this detour for O1–O7 and for all other H–O hydrogen bonds and O–O interactions is given by the values of $BPL - GBL - I$ in Tables 4–6. These O–O interactions with very unusual T-shaped bond-paths have very large values of $BPL - GBL - I$, that is, greater than 0.20 could be considered to be 'failed' hydrogen bonds, the authors are unaware of any similar interactions in the literature, although these T-shaped geometries are reminiscent of topologically unstable conflict structures associated with bifurcation

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catastrophe; any small perturbation would lead to the bond-path switching between the H and O atoms.[43]

(iii) The hydrogen bond: covalent character and cooperativity

In this work it can be seen from Tables 1 and 2 that the most energetically stable conformers of W_4 and W_5 contain the least number of hydrogen bonds, this corresponds to conformer *I* for W_4 and *PUCKERED_RING* for W_5 (see Fig. 1(a) and 2(a) respectively). This indicates the preference for planar (2-D_{QT}) structures for both the W_4 and W_5 conformer sets. Conversely, in Table 3 it can be seen that for W_6 the structure with the highest number of hydrogen bonds, the *PRISM*, is the most energetically stable. Previously, it was established that the hexamer W_6 is more energetically stable than both the tetramer[11] and the pentamer;[10] it was found that the $W_4/W_5/W_6$ relative energetic stability is not simply due to the larger number of water molecules in W_6 compared with W_5 and W_4 producing more hydrogen bonds.

A simple rule on the number of hydrogen bonds for structures obeying the Bernal–Fowler ice rules for both the W_4 and W_5 clusters: there can be n, $n + 1$ or $n + 2$ hydrogen bonds for W_n, $n = 4$ or 5, but the numbers of hydrogen bonds in W_6 can be n, $n + 1$, $n + 2$ or $n + 3$. The difference can be understood in terms of the topology of the W_4 and W_5 clusters preferring 2-D_{QT} geometries, whilst the most energetically stable W_6 clusters possess 3-D_{QT} geometries. Examination of Fig. 5 shows that there is a variation in the degree of covalent character, given by $H(r_{bcp}) < 0$ with bond-path length (BPL) for the three cluster sizes W_4–W_6. It can be seen that some of the conformers of the W_6 cluster set possess the highest degree of covalent character of any of the W_4–W_6 conformers. From Fig. 5(a) it can be seen that the value of $H(r_{bcp})$ of the 3-D_{QT} conformer *IX* (see Fig. 1(i)) is very atypical of the W_4 clusters, this is also true of the 3-D_{QT} *PYRAMID_5A–6B* conformers for W_5, although the difference is less dramatic. The values of $H(r_{bcp})$ of the 3-D_{QT} conformers, *PYRAMID_6A–6C*, are closer to the behavior of the majority of the W_6 clusters than the 3-D_{QT} conformers in W_4 clusters and W_5. This finding is in agreement with W_6 clusters being the transition from planar (2-D_{QT}) to compact (3-D_{QT}) structures.

IV. Conclusions

We have determined to make the definition of the geometry of molecules more consistent by using QTAIM instead of Euclidean geometry as the size regime where quantum mechanics is appropriate then a quantum mechanically consistent description of geometry should be used. The Poincaré–Hopf relation has been found to be useful for generating spanning sets of conformer topologies for the water clusters W_4, W_5 and W_6, where it has been used to generate 1, 2 and 3 new structures, with new topologies, for W_4, W_5 and W_6 respectively. It is most useful for 3-D_{QT} structures and less useful for 2-D_{QT} structures where there are no CCPs, it is not useful for linear (1-D_{QT}), chain-like structures without RCPs. There are degeneracies in the Poincaré–Hopf relation leading to different structures, as is clearly seen in Table 1. However, the energetic stability of the structures with the same solution to the Poincaré–Hopf relation can be compared using the QTAIM measures: ellipticity, ε_{H-O}, ε_{O-O}, $BPL - GBL - I$, D_{rc} and D_{rbO-O}. The positioning of the most energetically stable structure of W_6, the *PRISM*, in the middle of the topology phase diagram is consistent with earlier work stating that cluster energetic stability is not proportional to the number of hydrogen bonds.[9–11] In addition it was previously found[11] for W_6 that compact structures are energetically favored over non-compact cyclic structures by electronic energies with zero point energy corrections. The existence of O–O BCPs is not found to be associated with the global minima structure, but rather local minima or the structures at saddle points in the potential energy surface. The study of potential energy surfaces beyond just the global and local minima is of current interest and so the O–O BCPs provide a useful sensitive probe in real space to the potential energy surface. Also, structures obeying the Bernal–Fowler ice rules that contain topologically unstable O–O BCPs are associated with more energetically stable structures as compared with structures containing more topologically stable O–O BCPs. The increased energetic stability of the W_6 pyramid structures with higher numbers of O–O interactions reflects the preference for more complex 3-D_{QT} structures in the W_6 clusters and the O–O BCPs therefore, as a stabilizing mechanism compared with the W_5 and W_4 clusters.

Acknowledgements

The Knowledge Foundation (grant number 2004/0284) and the Hundred Talents Foundation of Hunan Province are gratefully acknowledged for the support of S.J. and S.R.K. J.D. was partially supported by Universidad Eafit, Internal grant number 173-000013. This work was made possible by the facilities of the Shared Hierarchical Academic Research Computing Network (SHARCNET: www.sharcnet.ca), through the kind auspices of our sponsor Dr P.W. Ayers, Department of Chemistry, McMaster University, Ontario, Canada.

References

1 M. D'Alessandro, A. Di Lella, M. Aschi, A. Di Nola and A. Amedei, *J. Mol. Liq.*, 2008, **142**, 111.
2 R. Ludwig, *Angew. Chem., Int. Ed.*, 2001, **40**, 1808.
3 P. Popelier, A. Lenstra, C. Van Alsenoy and H. Geise, *Struct. Chem.*, 1991, **2**, 3–9.
4 P. Popelier and R. Bader, *Chem. Phys. Lett.*, 1992, **189**, 542–548.
5 A. S. F Boyd, G. Cooke, F. M. A. Duclairoir and V. M. Rotello, *Tetrahedron Lett.*, 2003, **44**, 303.
6 A. S. F. Boyd, J. B. Carroll, G. Cooke, J. F. Garety, B. J. Jordan, S. Mabruk, G. Rosair and V. M. Rotello, *Chem. Commun.*, 2005, 2468.
7 G. Ferguson, C. D. Carroll, C. Glidewell, C. M. Zakaria and A. J. Lough, *Acta Crystallogr., Sect. B: Struct. Sci.*, 1995, **51**, 367–377.
8 A. Buckingham, J. Bene and S. McDowell, *Chem. Phys. Lett.*, 2008, **463**, 1.
9 H. Harker, M. Viant, M. Keutsch, E. Michael, R. McLaughlin and R. Saykally, *J. Phys. Chem. A*, 2005, **109**, 6483.
10 M. Brown, M. Keutsch and R. Saykally, *J. Chem. Phys.*, 1998, **109**, 9645.
11 M. Keutsch and R. Saykally, *Proc. Natl. Acad. Sci. U. S. A.*, 2001, **98**, 10533.
12 J. Gregory and D. Clary, *J. Phys. Chem. A*, 1997, **101**, 6813.
13 M. Tissandier, S. Singer and J. Coe, *J. Phys. Chem. A*, 2000, **104**, 752.
14 J. Pérez, C. Hadad and A. Restrepo, *Int. J. Quantum Chem.*, 2008, **108**, 1653.

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Published on 20 May 2011 on http://pubs.rsc.org | doi:10.1039/C1CP20135B

15 F. Ramírez, C. Hadad, D. Guerra, J. David and A. Restrepo, *Chem. Phys. Lett.*, 2011, **507**, 229.
16 G. Hincapié, N. Acelas, M. Castaño, J. David and A. Restrepo, *J. Phys. Chem. A*, 2010, **114**, 7809.
17 P. Dougord, J. Blanc, V. Bonačić-Koutecký, M. Broyer, J. Chevaleyre, J. Koutecký, J. Pittner, J. Wolf and L. Wöste, *Phys. Rev. Lett.*, 1991, **67**(19), 2638.
18 J. Blanc, V. Bonačic-Koutecký, M. Broyer, J. Chevaleyre, P. Dugourd, J. Koutecký, C. Scheuch, J. Wolf and L. Woste, *J. Chem. Phys.*, 1992, **96**, 1793.
19 B. Temelso and D. Sherrill, *J. Chem. Phys.*, 2005, **122**, 064315.
20 J. Pérez, E. Flórez, C. Hadad, P. Fuentealba and A. Restrepo, *J. Phys. Chem. A*, 2008, **112**, 5749.
21 M. Masella, N. Gresh and J. Flament, *J. Chem. Soc.*, 1998, **94**, 2745.
22 W. Koch and M. Holthausen, *Hydrogen bonds and weakly bound systems*, in *A Chemist's Guide to Density Functional Theory*, Wiley, New York, 2nd edn, 2002.
23 G. Chalasinski, M. Szczesniak, P. Cieplak and S. Scheiner, *J. Chem. Phys.*, 1991, **94**, 2873.
24 L. Ojamäe and K. Hermannsson, *J. Phys. Chem.*, 1994, **98**, 4271.
25 M. Masella and J. Flament, *J. Chem. Phys.*, 1998, **108**, 7141.
26 B. King and F. Weinhold, *J. Chem. Phys.*, 1995, **103**, 333.
27 J. David, D. Guerra and A. Restrepo, *J. Phys. Chem. A*, 2009, **113**, 10167.
28 J. Murillo, J. David and A. Restrepo, *Phys. Chem. Chem. Phys.*, 2010, **12**, 10963.
29 R. F. W. Bader, *Atoms in Molecules: a Quantum Theory*, Clarendon, Oxford, 1994.
30 S. Jenkins and I. Morrison, *Chem. Phys. Lett.*, 2000, **317**, 97.
31 N. Castillo, C. F. Matta and R. J. Boyd, *Chem. Phys. Lett.*, 2005, **409**, 265–269.
32 R. F. W. Bader, *J. Phys. Chem. A*, 2009, **113**, 10391–10396.
33 P. W. Ayers and S. Jenkins, *J. Chem. Phys.*, 2009, **130**(15), 154104.
34 S. Jenkins, S. R. Kirk, A. S. Cote, D. K. Ross and I. Morrison, *Can. J. Phys.*, 2003, **81**, 225.
35 S. Jenkins and I. Morrison, *J. Phys.: Condens. Matter*, 9207, **13**, 2001.
36 S. Jenkins, S. R. Kirk and P. W. Ayers, in *Physics and Chemistry of Ice*, ed. W. F. Kuhs, Royal Society of Chemistry, Cambridge, 2007, p. 273.
37 S. Jenkins, S. R. Kirk and P. W. Ayers, in *Physics and Chemistry of Ice*, ed. W. F. Kuhs, Royal Society of Chemistry, Cambridge, 2007, p. 249.
38 D. Cremer and E. Kraka, *Croat. Chem. Acta*, 1984, **57**, 1250.
39 E. D. Isaacs, A. Shukla, P. M. Platzman, D. R. Hamann, B. Barbiellini and C. A. Tulk, *Phys. Rev. Lett.*, 1999, **82**(3), 600.
40 Todd A. Keith, *AIMAll (Version 10.09.12)*, 2010 (aim.tkgristmill.com).
41 J. D. Bernal and R. H. Fowler, *J. Chem. Phys.*, 1933, **1**, 515.
42 R. M. Shields, B. Temelso, K. A. Archer, T. E. Morrell and G. C. Shields, *J. Phys. Chem. A*, 2010, **114**, 11725–11737.
43 L. J. Farrugia, C. Evans, D. Lentz and M. Roemer, *J. Am. Chem. Soc.*, 2009, **131**, 1251–1268.
44 S. Jenkins, C. Rong, S. R. Kirk, D. Yin and S. Liu, *J. Phys. Chem. A*, 2011, DOI: 10.1021/jp202294n. Publication Date (Web): May 10, 2011.

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